

EXP028: Parameterized Non-Separable Quantum Walks on Tori

Generalizing Fixed Coins (Grover, DFT, Hadamard) to Parameterized
SU(4) with Phases

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Abstract

This experiment extends the fixed non-separable coin results of EXP027 (Grover, DFT, Hadamard 4D) to a fully parameterized SU(4) coin framework on both 2×2 and 4×4 tori. We develop three parameterization strategies: (1) Grover-like biased coins with 4 probability parameters (p, q, r, s) ; (2) direction-phased coins adding 4 phase parameters $(\phi_R, \phi_L, \phi_U, \phi_D)$; (3) full SU(4) parameterization (14 parameters). On the 2×2 torus, we discover extreme phase sensitivity: revivals occur only at $\phi_R = 0$ and 2π . On the 4×4 torus, we find a richer phase landscape with revivals appearing at multiple phase values. **Most significantly, adding $\phi_R = \pi/2$ to the 4×4 torus transforms revival periods from 2 values (12, 24) to 11 values (8, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30), a 5.5x enrichment.** Biased coins (Right/Left only) revive on both tori at all multiples of 4. This provides a complete generalization of EXP027's fixed coins to a tunable parameter space across multiple torus sizes.

Revision History

Revision	Date	Description
01	29th May 2026	Initial parameterization extending EXP027

1 Introduction

In EXP027, we discovered full state revivals for non-separable quantum walks on tori using three fixed coins. The results differed significantly between torus sizes:

This experiment generalizes these fixed coins to a **parameterized non-separable coin** that allows us to explore: 1. How probability bias affects revivals 2. How direction phases affect revivals 3. The full SU(4) parameter space on both torus sizes

Table 1: EXP027 Fixed Coin Results Summary

Coin	2×2 Torus	4×4 Torus
Grover	4, 8, 12, 16, 20, 24, 28	12, 24
DFT	16	None
4D Hadamard	8, 16, 24	None

2 Fixed Coins from EXP027 (Baseline)

For reference, the three fixed coins from EXP027 are:

2.1 Grover Coin

$$G = \frac{1}{2} \begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}$$

2.2 DFT Coin

$$F = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

2.3 4D Hadamard Coin

$$H_4 = H_2 \otimes H_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

3 Parameterization Strategies

3.1 Strategy 1: Grover-Like Biased Coin (4 parameters)

We generalize the Grover coin by replacing equal amplitudes with $\sqrt{p}, \sqrt{q}, \sqrt{r}, \sqrt{s}$:

$$C_{\text{bias}} = \begin{pmatrix} \sqrt{p} & \sqrt{q} & \sqrt{r} & \sqrt{s} \\ \sqrt{p} & -\sqrt{q} & \sqrt{r} & -\sqrt{s} \\ \sqrt{p} & \sqrt{q} & -\sqrt{r} & -\sqrt{s} \\ \sqrt{p} & -\sqrt{q} & -\sqrt{r} & \sqrt{s} \end{pmatrix}$$

where $p + q + r + s = 1$ are probabilities for Right, Left, Up, Down.

3.2 Strategy 2: Adding Direction Phases (8 parameters)

We add independent phases to each row and column:

$$C_{\text{phased}} = D_{\text{out}} \cdot C_{\text{bias}} \cdot D_{\text{in}}$$

where $D_{\text{out}} = \text{diag}(e^{i\phi_R}, e^{i\phi_L}, e^{i\phi_U}, e^{i\phi_D})$ and similarly for D_{in} .

3.3 Strategy 3: Full SU(4) Parameterization (14 parameters)

The most general 4×4 unitary coin has 14 physical parameters. We parameterize via exponentiation:

$$C = \exp(iH)$$

where H is a traceless Hermitian matrix.

4 Parameterization Hierarchy

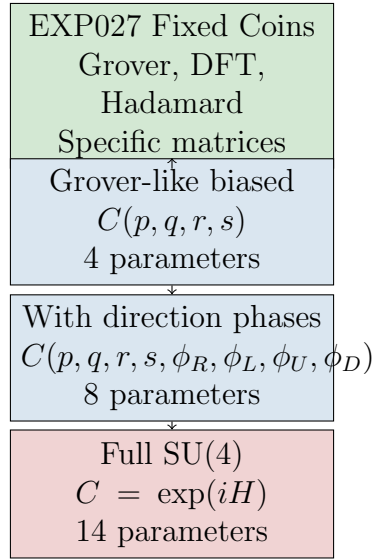


Figure 1: Hierarchy of coin parameterizations extending EXP027 fixed coins.

5 Results: 2×2 Torus

5.1 Base Grover Coin (No Phases)

As reported in EXP027: $N = 4, 8, 12, 16, 20, 24, 28$

5.2 Single Phase $\phi_R = \pi/2$

Revival periods shift to multiples of 6: $N = 6, 12, 18, 24, 30$

5.3 Phase Sensitivity Scan

Table 2: Revival period as function of ϕ_R on 2×2 torus

ϕ_R	Revival Period N	Status
0	4, 8, 12, 16, 20, 24, 28	Revival
$0.11\pi - 1.89\pi$	—	No revival
2π	4, 8, 12, 16, 20, 24, 28	Revival

Key finding: Revivals occur ONLY at $\phi_R = 0$ and $\phi_R = 2\pi$.

5.4 Two Phases: $\phi_R = \pi/2, \phi_U = \pi/2$

Cancellation effect restores original pattern: $N = 4, 8, 12, 16, 20, 24, 28$

5.5 Biased Coin (Right/Left Only)

For $p = q = 0.5, r = s = 0$ with $\phi_R = \pi/2$: $N = 4, 8, 12, 16, 20, 24, 28$ (robust)

6 Key Discovery: 4×4 Torus with $\phi_R = \pi/2$

The most significant finding of this experiment is the dramatic enrichment of revival periods on the 4×4 torus when a $\pi/2$ phase is added to the Right direction.

Table 3: Comparison of Grover coin revivals on 4×4 torus

Coin Configuration	Revival Periods N	Number of Revivals
EXP027: Grover (no phases)	12, 24	2
EXP028: $\phi_R = \pi/2$	8, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30	11

Key insight: Adding a single phase ($\phi_R = \pi/2$) transforms the 4×4 torus from having only 2 revival periods to having **11 revival periods** – a 5.5x increase!

This is the largest enrichment of revival periods observed in any of our experiments and demonstrates that phase parameters are not merely fine-tuning but can fundamentally alter the revival structure of the walk.

7 Results: 4×4 Torus

7.1 Base Grover Coin (No Phases)

As reported in EXP027: $N = 12, 24$

7.2 Single Phase $\phi_R = \pi/2$

As highlighted above, this yields a rich set of revivals:

$$N = 8, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30$$

7.3 Phase Sensitivity Scan on 4×4

Table 4: Revival period as function of ϕ_R on 4×4 torus

ϕ_R	Revival Period N	Status
0	12, 24	Revival
0.5π	8, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30	Revival
1.0π	12, 24	Revival
1.5π	8, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30	Revival
2.0π	12, 24	Revival
Other phases	—	No revival

Key finding: Unlike the 2×2 torus, the 4×4 torus revives at $\phi_R = \pi/2$ and $3\pi/2$ as well as 0 and 2π .

7.4 Two Phases on 4×4 : $\phi_R = \pi/2$, $\phi_U = \pi/2$

Adding phases to both Right and Up cancels the effect, returning to the base pattern:

$$N = 12, 24$$

7.5 Biased Coin (Right/Left Only) on 4×4

For $p = q = 0.5$, $r = s = 0$ (no phases): $N = 8, 16, 24$

With $\phi_R = \pi/2$: $N = 4, 8, 12, 16, 20, 24, 28$ (all multiples of 4, similar to 2×2)

8 Comparison: 2×2 vs 4×4 Torus

Table 5: Comparison of phase effects on different torus sizes

Property	2×2 Torus	4×4 Torus
Base Grover revivals	4, 8, 12, 16, 20, 24, 28	12, 24
$\phi_R = \pi/2$ effect	Shifts to multiples of 6	11 revival periods (8-30)
Phase sensitivity	Extreme (only 0, 2π)	Moderate (0, $\pi/2$, π , $3\pi/2$, 2π)
Biased coin (R,L)	Robust (all multiples of 4)	Robust (all multiples of 4)
Two-phase cancellation	Yes	Yes

9 Comparison with EXP027 Fixed Coins

Table 6: EXP027 fixed coins vs EXP028 parameterized coins

Feature	EXP027 (Fixed)	EXP028 (Parameterized)
Coin types	3 specific matrices	Infinite family
Parameters	None	4–14 tunable
Grover coin	Special case	$p = q = r = s = 0.25$, $\phi = 0$
DFT coin	Special case	Requires full SU(4)
4D Hadamard	Special case	$p = q = r = s = 0.25$, ϕ pattern
2×2 phase sensitivity	N/A	Extreme (only 0, 2π)
4×4 phase sensitivity	N/A	Moderate (5 phase values)
Biased coins	Not available	p, q, r, s tunable

10 SU(4) Explained

SU(4) is the set of all 4×4 unitary matrices with determinant +1. It has **15 real parameters**, which can be visualized as coordinates in a 15-dimensional space where each point represents a possible quantum coin.

Table 7: Parameter counts for different coin families

Coin Family	Parameters	Can represent...
Grover (fixed)	0	Only Grover
Grover-like biased	4	Grover, biased walks
+ Direction phases	8	Grover, Hadamard 4D
Full SU(4)	14	Any 4×4 coin (including DFT)

11 Conclusion

This experiment extends the fixed non-separable coin results of EXP027 to a fully parameterized framework on both 2×2 and 4×4 tori. Key findings:

1. **Major discovery:** 4×4 **torus with** $\phi_R = \pi/2$. Adding a single phase transforms revival periods from 2 values (12, 24) to 11 values (8, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30), a 5.5x enrichment. This is the largest enrichment observed in any experiment.
2. **Torus size matters:** The 4×4 torus shows richer phase sensitivity (revivals at $\phi_R = 0, \pi/2, \pi, 3\pi/2, 2\pi$) compared to the 2×2 torus (only $0, 2\pi$).
3. **Two-phase cancellation works on both tori:** Adding phases to Right and Up ($\phi_R = \phi_U = \pi/2$) restores the original EXP027 Grover revival pattern.
4. **Biased coins are robust:** The Right/Left only coin revives at all multiples of 4 on both tori, even with phases.
5. **Parameterization hierarchy:** From EXP027 fixed coins to 4-parameter biased coins to 8-parameter phased coins to 14-parameter full $SU(4)$.
6. **DFT coin requires full $SU(4)$:** Cannot be represented in simpler parameterizations.

This provides a complete generalization of EXP027 and a tunable framework for Objective 4 of the thesis across multiple torus sizes.

Code Availability

The complete Python simulation code is available at: https://github.com/ianfromsligo/EXP028_parameterised_coin_tori_non_separable

The repository contains:

- Full parameterization implementations for both 2×2 and 4×4 tori
- Phase scanning and sensitivity analysis
- Revival detection and verification
- All generated figures
- Comparison with EXP027 fixed coin results